Education and Annual Earnings in Massachusetts, 2024

ECON 3916 Midterm Project | November 2026

Repository: https://github.com/rohit-jain8/econ3916-education-earnings

Notebook: https://colab.research.google.com/drive/1lhMf-s9C-BmkkWNxUKEhS1AplfIzkrh_?usp=sharing

	Project at a glance
Question	How is education associated with annual earnings among Massachusetts adults ages 25–64 with positive earnings in 2024, controlling for age and sex?
Hypothesis	Adults with higher education levels will generally have higher annual earnings.
Headline finding	Compared with high school/GED, bachelor's and graduate degrees are associated with approximately 99% and 152% higher annual earnings, respectively, holding age and sex fixed.
Data	2024 ACS 1-Year Massachusetts person-level PUMS; 31,442 adults in the analysis sample.
Method	OLS of log annual earnings on education categories, age, age squared, and sex; HC3 robust standard errors, plus weighted and full-time comparisons.

1. The Question and the Data

Higher education is often viewed as a way to improve career opportunities and earning potential. I wanted to see how large the earnings differences actually are in recent Massachusetts data, and whether they remain after accounting for age and sex. My question is about an association, not whether schooling itself causes higher earnings.

I use the U.S. Census Bureau's 2024 American Community Survey (ACS) 1-Year Public Use Microdata Sample (PUMS) for Massachusetts. Each row represents one person. The original file has 74,617 people and 286 variables. I keep adults ages 25–64 with positive annual earnings (PERNP), leaving 31,442 observations. Positive earnings are required because the outcome is ln(PERNP); this also means my results do not describe people without earnings.

My main explanatory variable is educational attainment (SCHL), grouped into less than high school, high school/GED, some college or associate degree, bachelor's, and graduate education. High school/GED is the reference group. I control for age, age squared, and sex. Age squared allows earnings to rise more slowly or decline later in a career. Education cannot change age or sex, so

these are reasonable controls. I leave occupation and hours worked out of the main model because education may affect both; including them would change the relationship being measured.

The original data contain 10,609 missing PERNP values and 1,745 missing SCHL values; age, sex, and person weights (PWGTP) have none. Some gaps may arise because questions do not apply to certain people, but I cannot establish whether all missingness is MAR or MNAR. The final analysis variables have no missing values after the sample restrictions. All variables refer to individuals in Massachusetts in 2024, so no policy pre-period is needed.

Table 1. Sample selection	**People**
Original Massachusetts person file	74,617
Age 25–64, positive earnings, usable education and controls	31,442

Note. The analysis restriction combines age, positive earnings, and complete model variables; it is not a step-by-step attrition decomposition.

Table 2. Summary statistics	**Mean**	**Median**	**SD**
Annual earnings ($)	97,782	72,000	104,392
Age (years)	43.99	44	11.62
Usual hours per week	39.76	40	11.11

Note. Unweighted descriptive statistics for N = 31,442. Earnings range from $20 to $1,186,000.

The sample restrictions matter for interpretation. Someone who did not work during 2024 is not included, even if they completed a degree and might earn income in another year. Similarly, the results describe annual earnings among people with earnings, not the probability of having a job. These are related questions, but they would require a different outcome and sample. The person weights are available for the ACS data, so I use them in an additional model rather than assuming that an unweighted sample automatically represents the state population.

2. Descriptive Evidence

Annual earnings are strongly right-skewed (Figure 1). The median is $72,000, compared with a mean of about $97,782. A small group of high earners stretches the distribution to the right. I cap the figure at the 99th percentile for display only, keeping all observations in the regressions. This is one reason I use log earnings as my outcome.

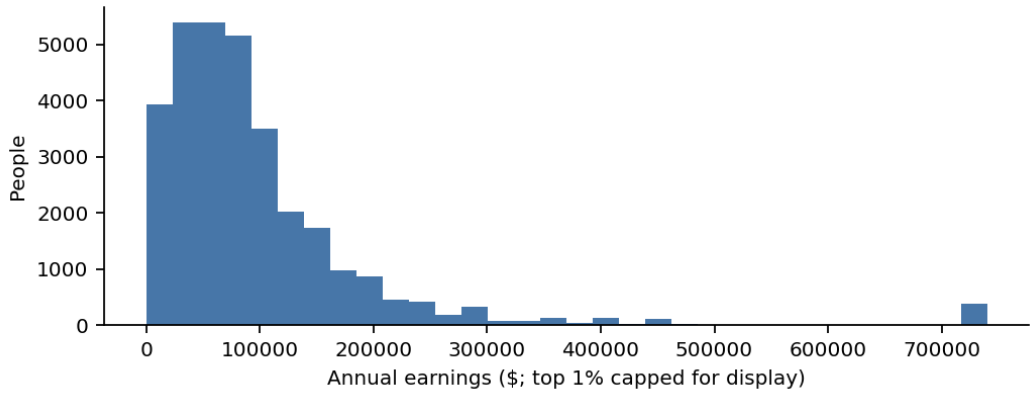


Figure 1. Annual earnings are right-skewed; the top 1% is capped for display only.

Median earnings rise steadily with education (Figure 2): $37,650 for less than high school, $47,500 for high school/GED, $55,000 for some college or an associate degree, $85,000 for a bachelor's degree, and $100,000 for graduate education. The bachelor's median is $37,500 above the high school/GED median. These comparisons do not yet control for age, sex, or differences in hours and weeks worked.

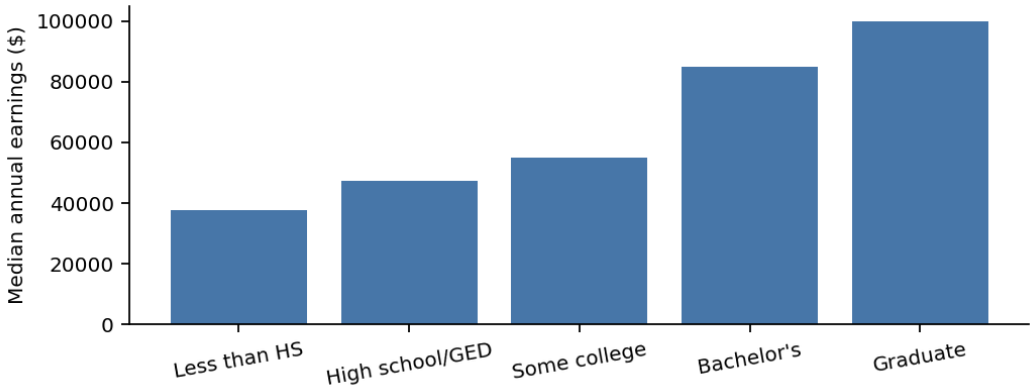


Figure 2. Median annual earnings rise across the five education groups (unweighted).

The descriptive pattern is especially noticeable between some college and a bachelor's degree: median annual earnings increase by $30,000 across those two categories. The gap from bachelor's to graduate education is another $15,000. These are comparisons of group medians, not the additional earnings that the same person would necessarily receive by earning another degree. The groups differ in age, occupation, and how much they work, which is why I next use a regression with controls.

3. The Model

I estimate OLS with HC3 robust standard errors. The dependent variable is log annual earnings; the main explanatory variables are four education indicators, with high school/GED omitted as the reference. The model also includes age, age squared, and a female indicator:

$$\ln(\text{Earnings}_i) = \beta_0 + \beta_1 \text{LessHS}_i + \beta_2 \text{SomeCollege}_i + \beta_3 \text{Bachelors}_i + \beta_4 \text{Graduate}_i + \beta_5 \text{Age}_i + \beta_6 \text{Age}^2_i + \beta_7 \text{Female}_i + u_i$$

The main model uses 31,442 observations and has $R^2 = 0.171$, meaning these variables explain about 17.1% of the variation in log annual earnings. Table A1 in the appendix gives the full regression results. The bachelor's coefficient is 0.6896 (HC3 SE 0.0172). Converting the log coefficient using $100 \times (e^{\beta} - 1)$, that is about 99.3% higher annual earnings than high school/GED, holding age and sex constant. The graduate coefficient is 0.9248 (HC3 SE 0.0173), or about 152.1% higher earnings. Both estimates are statistically significant at the 1% level.

The other education estimates follow the same pattern: some college/associate education is associated with approximately 17.9% higher earnings, while less than high school is associated with approximately 30.6% lower earnings, relative to high school/GED. The age coefficient is positive (0.0960) and age squared is negative (−0.000974), suggesting that earnings initially rise with age but at a slowing rate. The female coefficient is −0.3863, indicating a conditional earnings difference that may also reflect unmeasured work patterns and occupation.

Following the checkpoint feedback, I also estimated a model using ACS person weights (PWGTP). The bachelor's and graduate coefficients are 0.6527 and 0.8831, respectively, still positive and similar in magnitude. In a separate sample of 23,633 full-time, year-round workers (at least 35 usual hours per week and 50 weeks per year), the corresponding coefficients are 0.5895 and 0.7870. The smaller estimates are consistent with work schedules explaining part of the annual earnings difference, but the sample also changes, so this is not a clean measure of that mechanism. Table A1 reports all three specifications.

I interpret these coefficients as comparisons with the high school/GED group, not as a prediction that every person with a bachelor's or graduate degree earns the same amount. Because the dependent variable is in logs, the coefficients describe proportional differences, and the exact percentage conversion matters when the coefficients are large. The estimates are also quite precise statistically, as shown by the small robust standard errors, but statistical precision does not rule out omitted-variable bias. The weighted and full-time estimates are useful checks of how the answer changes under different sample definitions and weighting choices, rather than proof that the main coefficient is causal.

4. Diagnostics

I used the Topic 14 diagnostics to examine the main OLS model. Figure A1 in the appendix plots residuals against fitted log earnings and across education categories. These plots help check whether the spread of the errors changes and whether there are noticeable patterns not captured by the model.

The residuals-versus-fitted-values plot shows that the spread of the errors changes across predicted earnings levels, which is consistent with the heteroskedasticity identified by the Breusch–Pagan test. The residuals also vary across education groups, although there is no obvious pattern suggesting that one education category alone explains the issue.

The Breusch–Pagan test rejects constant error variance ($p = 8.12 \times 10^{-21}$). Heteroskedasticity does not by itself bias the OLS coefficient estimates, but conventional standard errors may be unreliable. I therefore report HC3 robust standard errors for inference throughout the main results.

The education variables have VIFs of roughly 1.21 to 2.00, so multicollinearity is not a major concern for the comparisons I care most about. Age and age squared have much larger VIFs, around 79, because one is calculated from the other. I keep both to allow a curved age–earnings relationship. Cook's distance flags 1,609 observations above the screening cutoff 4/n; the largest value is about 0.00560. That cutoff is a prompt to inspect observations, not a reason to delete them automatically. I retain them in the main analysis. Table A2 summarizes the diagnostic results.

The diagnostics answer different questions. The Breusch–Pagan result tells me the usual constant-variance assumption is not supported, so robust inference is appropriate. VIF concerns whether predictors overlap enough to make individual estimates unstable; the main education comparisons have much lower VIFs than the age terms. Cook's distance instead looks at whether individual observations have unusual influence. None of these tests can tell me whether unobserved family background or ability is affecting the estimated education relationship. That is a separate limitation of the research design.

5. Threats and Conclusion

The main limitation is omitted variable bias. Ability, family background, financial resources, work experience, and career choices could affect both education and earnings. My controls do not account for all of these factors, so the coefficients should not be interpreted as the causal return to education. The outcome also measures annual earnings rather than hourly pay: people may earn more partly because they work more hours or weeks. The full-time comparison helps explore this issue but does not eliminate it.

The results apply to Massachusetts adults ages 25–64 with positive earnings in 2024, not to everyone in the state or to other years. The weighted model helps assess how the findings change when using ACS person weights, but a full survey-design analysis would require additional treatment of the sampling design. HC3 standard errors address heteroskedasticity for the fitted model, not these broader identification concerns.

Overall, education is strongly associated with annual earnings in this sample. Relative to high school/GED, bachelor's and graduate education are associated with approximately 99% and 152%

higher earnings after controlling for age and sex. The positive pattern remains in the weighted and full-time comparisons, although the coefficients are smaller. I would next examine work experience and hours worked as possible explanations, while recognizing that adding variables affected by education changes what the education coefficient measures.

References

U.S. Census Bureau. (2024). American Community Survey, 2024 1-Year Public Use Microdata Sample, Massachusetts person-level data. https://www2.census.gov/programs-surveys/acs/data/pums/2024/1-Year/

Appendix

Table A1. Log annual earnings regressed on education and controls

Variable	Main OLS	Weighted	Full-time
Less than high school	-0.3656	-0.3533	-0.2469
	(0.0365)	(0.0463)	(0.0317)
Some college / associate	0.1642	0.1333	0.1238
	(0.0193)	(0.0246)	(0.0136)
Bachelor's	0.6896	0.6527	0.5895
	(0.0172)	(0.0213)	(0.0126)
Graduate	0.9248	0.8831	0.7870
	(0.0173)	(0.0221)	(0.0128)
Age	0.0960	0.0901	0.0744
	(0.0044)	(0.0055)	(0.0032)
Age squared	−0.000974	-0.0009	-0.0007
	(0.000049)	(0.0001)	(0.0000)
Female	-0.3863	-0.3828	-0.2493
	(0.0112)	(0.0143)	(0.0085)
Constant	8.5373	8.6842	9.2344
	(0.0928)	(0.1153)	(0.0671)
N	31,442	31,442	23,633
R^2	0.171	—	—

Note. Outcome: ln(annual earnings). Reference: high school/GED. HC3 robust standard errors in parentheses. Weighted specification uses ACS PWGTP; full-time means ≥35 usual hours/week and ≥50 weeks/year. R^2 is reported for the main model only.

Table A2. Main-model diagnostics

Diagnostic	Result
Breusch–Pagan p-value	8.12×10^{-21}
VIF: education indicators	1.21–2.00
VIF: age / age squared	about 79.21 / 79.19
VIF: female	1.01
Maximum Cook's distance	0.00560
Observations above 4/n	1,609

Figure A1. Main-model residual plots

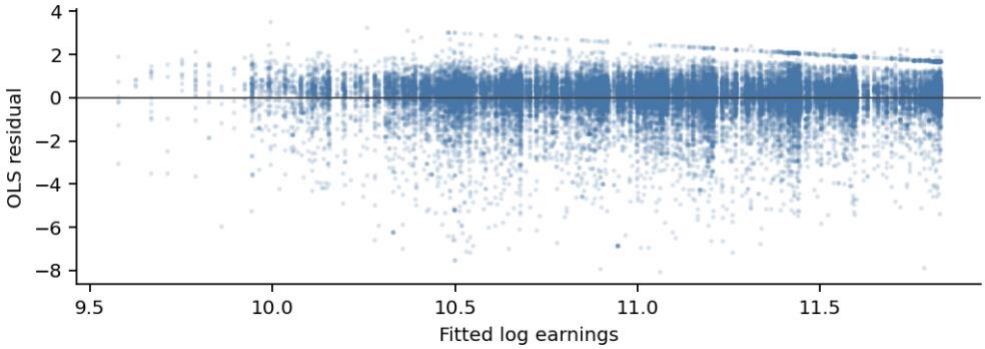


Figure A1a. Residuals against fitted log annual earnings.

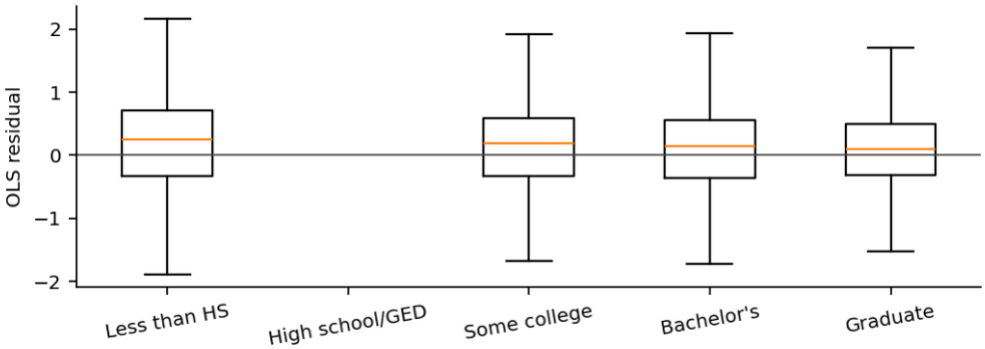


Figure A1b. Residuals across education categories (outliers hidden for readability).
Reproducibility: The executed notebook and project dataset generate the reported regression and diagnostic results.